

ZHENZHUO LI

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EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) BSc, Data Science and Big Data Technology Sep 2023 – Jun 2027 Expected

- CGA: 3.807/4.3; Major CGA: 3.837/4.3; Dean's List for three semesters
- Relevant Coursework: Advanced Algorithms and Data Structures (A+), Introduction to AI (A+), Database Management Systems (A+), Linear Algebra (A+), Design and Analysis of Algorithms (A), Machine Learning (A-), Applied Statistics (A-), Cloud Computing and Big Data Systems (A-)

Osaka University Exchange Student Fall 2025

- Graduate School exchange program, Japan

RESEARCH INTERESTS

Time-Series Representation Learning; Deep Learning for Sequential Data; Transfer and Adaptation under Distribution Shift; Reliable Machine Learning; Online Learning and Sequential Decision-Making; Data-Centric AI, with applications to financial and other complex real-world systems.

PUBLICATIONS

- Yuanjian Xu, Jianing Hao, Anxian Liu, **Zhenzhuo Li**, Shichang Meng, Siyuan Yuan, and Guang Zhang, "LENS: Large Pre-trained Transformer for Exploring Financial Time Series Regularities," in *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF 2025)*, pp. 771–778, 2025. DOI: 10.1145/3768292.3770349.

RESEARCH EXPERIENCE

LENS: Large Pre-trained Transformer for Financial Time Series Undergraduate Research Assistant / Co-author Jun 2024 – Oct 2024

Supervisor: Prof. Guang Zhang, HKUST(GZ)

- Developed parallel and memory-efficient preprocessing pipelines for terabyte-scale financial time-series data, supporting Transformer pretraining through chunk-based cleaning, aggregation, and model-input construction.
- Achieved up to $13\times$ faster data cleaning and $6\times$ faster statistical aggregation in internal benchmarks by restructuring data access and parallel processing.
- Contributed data-processing and experimental infrastructure to the resulting paper published at ACM ICAIF 2025.

Xuanyuan Private Fund Management (Guangdong) Co., Ltd. Quantitative Research Intern Jul 2026 – Present

Guangzhou, China

- Conducted end-to-end research on neural feature and factor discovery from Level-2 equity-market data, covering data construction, model development, signal evaluation, and decision-oriented analysis.
- Built scalable pipelines for multi-year, market-wide minute-level panels, including temporal and cross-asset alignment, memory-mapped storage, parallel I/O, sample filtering, and model-ready dataset construction.
- Developed leakage-aware evaluation procedures for intraday and cross-day signals, including information-coefficient analysis, temporal-stability diagnostics, entry-timing evaluation, and sample-selection audits.
- Investigated tree-based and neural sequence models, including LightGBM, XGBoost, and GRU, for market-exposure and intraday-entry decisions under non-stationary conditions.

SpatialReflect: Spatially Disentangled Reflection for Controllable Generation First-author Independent Research Jun 2025 – May 2026

Supervisor: Prof. Zeke Xie, HKUST(GZ)

- Formulated failure modes in multi-region diffusion-based generation and developed a training-free framework for spatially controlled synthesis.
- Designed controlled experiments and completed a full first-author manuscript, gaining end-to-end experience in problem formulation, method development, implementation, evaluation, and scientific writing.

- Developed an early leakage-aware factor-evaluation prototype that was later extended through institutional Level-2 data research.

SELECTED PROFESSIONAL EXPERIENCE

Shanghai Yunyan Private Fund Management Co., Ltd. AI Investment Research Intern May 2026 – Jul 2026

- Conducted AI-assisted research on semiconductor and AI-infrastructure companies by synthesizing company disclosures, industry research, expert commentary, and open-source information.
- Explored LLM-agent workflows for information screening and multi-source verification, motivating an interest in reliable AI systems operating under noisy and dynamically changing information.

SELECTED AWARDS & HONORS

- ASC World University Supercomputer Competition 2026 — **First Prize**, Team Member
- HKUST(GZ) TradeMaster Cup Quantitative Trading Competition 2026 — **5th Place**, Team Leader
- INFH Lizhi Innovation Talent Scholarship, 2025
- MCM/ICM Mathematical Contest in Modeling 2025 — Honorable Mention
- ASC World University Supercomputer Competition 2025 — Second Prize, Team Member
- Dean's List — Three Semesters
- Admission Scholarship, First Class, HKUST(GZ), 2023

TECHNICAL SKILLS

Programming and Research Engineering: Python, C++, SQL; Linux, Git, Slurm, CUDA, MPI; scalable data preprocessing, parallel I/O, memory-mapped arrays, experiment automation, and reproducible research pipelines

Deep Learning: PyTorch; implementation, training, and evaluation of Transformer-, GRU-, and diffusion-based models; custom data pipelines, masking and normalization, ablation studies, and training diagnostics

Representation and Transfer Learning: pretraining and fine-tuning workflows, reuse of learned representations across tasks and domains, frozen-versus-adapted model comparisons, domain-shift evaluation, and adaptation of pretrained models to specialized sequential or generative tasks

Time-Series and Reliable ML: multivariate sequence modeling, temporal and cross-sectional evaluation, leakage-aware dataset construction, walk-forward validation, distribution-shift analysis, stability diagnostics, positive controls, and decision-oriented evaluation

Machine Learning: LightGBM, XGBoost, scikit-learn; supervised learning, feature engineering, model selection, hyperparameter tuning, and error analysis

Reinforcement Learning and Sequential Decision-Making: systematic coursework (Introduction to Reinforcement Learning) with source-level reimplementations of bandit, tabular, and deep RL algorithms (UCB, Q-learning, DQN, REINFORCE, A2C, DPG); MDP and bandit modeling, Bellman equations, on/off-policy and model-based/free analysis, experience replay and target networks, actor-critic training, and importance sampling; literature research on distributional RL for risk-sensitive decision problems

LANGUAGES

Chinese: Native; English: Proficient, undergraduate instruction in English; Japanese: JLPT N1